⊙ Riemann-Roch Space $L(D)$.

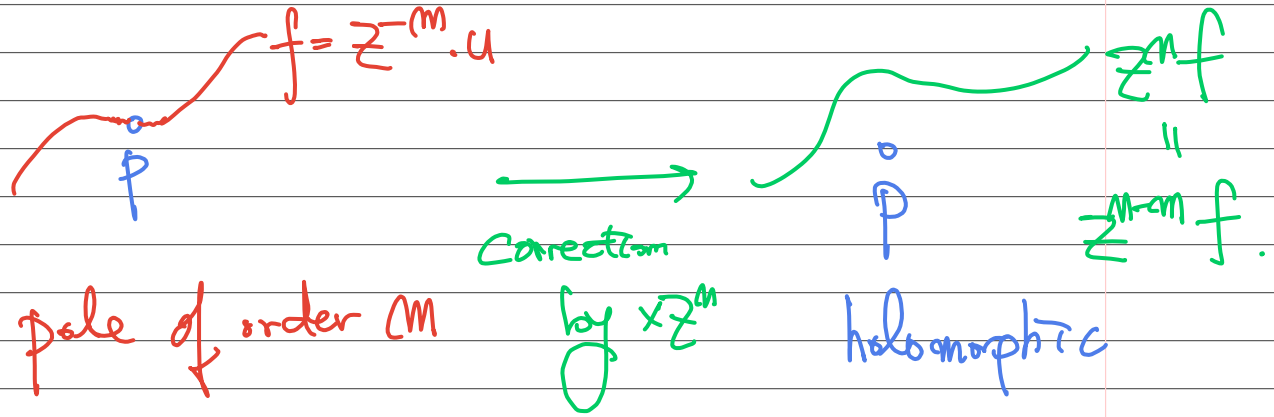
Divisor Group

$$\mathrm{Div}(X) = \Big\{ \sum_{p \in X} n_p [p] : n_p \in \mathbb{Z}, \; n_p = 0 \text{ for all but finitely many } p \Big\}.$$

$$D = \sum_{p \in X} n_p [p] \geq 0 \overset{\text{def}}{\Leftrightarrow} \forall p \in X, \; n_p \geq 0.$$

$$f \in M(X)^\times \xrightarrow{z(p)=0} f(z) = z^m u(z) \text{ with } u(p) \neq 0.$$

We want to allow at MOST a pole of order n at $p \in X$



2

$$\mathrm{ord}_p(f) = \mathrm{ord}_p(z^{-m}\cdot u) = -m$$

$$\mathrm{ord}_p(z^n f) = n + \mathrm{ord}_p(f) \geq 0$$

(in order to holomorphic)

$$\therefore\ f = z^{-m}\cdot u \quad \xrightarrow{\text{Correction}} \quad z^n f = z^{n-m} f$$

$$= z^{-\mathrm{ord}_p(m)}\cdot u \qquad\qquad = z^{n+\mathrm{ord}_p(f)}\cdot f$$

f has pole of at most n at p.

$\Updownarrow$

$z^n f$ is holomorphic at p

$\Updownarrow$

$$n + \mathrm{ord}_p(f) \geq 0.$$

$$D = \sum_p n_p [p] \qquad\qquad \mathrm{div}(f) = \sum_p \mathrm{ord}_p(f)\cdot[p]$$

$$\begin{cases} \mathrm{div}(z) = [0] - [\infty] \\ \mathrm{div}(1/z) = [\infty] - [0] \end{cases}$$

Define

$$L(D) := \{ f \in M(X)^\times : \operatorname{div}(f) + D \geq 0 \} \cup \{0\}$$

$\rightsquigarrow$ $L(D)$ is a vector space over $\mathbb{C}$.

$$\ell(D) := \dim_{\mathbb{C}} L(D)$$

Rmk. $f \in L(D) \Leftrightarrow \operatorname{div}(f) + D \geq 0$

$\Leftrightarrow \forall p \in X,\ \operatorname{ord}_p(f) + n_p \geq 0$

$\Leftrightarrow \forall p \in X,\ \operatorname{ord}_p(f) \geq -n_p$

Eg. Take $D = n[\infty]$ on $\mathbb{CP}^1$.

$$L(n[\infty]) = \{ f \in M(\mathbb{CP}^1)^\times : \operatorname{div}(f) + n[\infty] \geq 0 \} \cup \{0\}$$

$$\operatorname{div}(f) + n[\infty] = \sum_{p \in \mathbb{CP}^1} \operatorname{ord}_p(f)[p] + n[\infty]$$

$\mathbb{CP}^1 \hookrightarrow \simeq \mathbb{C} \cup \{\infty\}$

$$= \sum_{p \in \mathbb{C}} \operatorname{ord}_p(f)[p] + (\operatorname{ord}_\infty(f) + n)[\infty]$$

4

$\therefore \operatorname{div}(f) + n[\infty] \geq 0$

$\Leftrightarrow \begin{cases} \operatorname{ord}_p(f) \geq 0 & \text{if } p \neq \infty \quad \cdots (*) \\ \operatorname{ord}_\infty(f) + n \geq 0. & \cdots (**) \end{cases}$

$\therefore L(n[\infty]) = \{$ meromorphic functions on $\mathbb{CP}^1$ with no poles except at ∞, (by $(*)$) and there of order at most n. $\}$ (by $(**)$)

Eg. - (1) $D = 0$ (zero divisor)

$L(0) = \{ f \in M(\mathbb{CP}^1)^\times : \operatorname{div}(f) \geq 0 \} \cup \{0\}$

$\rightsquigarrow \forall p \in \mathbb{CP}^1, \ \operatorname{ord}_p(f) \geq 0$

$\therefore L(0) = \{$ holomorphic functions on C.R.S $\mathbb{CP}^1 \}$

$= \{$ constant $\} = \mathbb{C}$

$\therefore \ell(0) = \dim_{\mathbb{C}} L(0) = 1.$

5

- (2) $D = [0]$

$$\operatorname{div}(f) + [0] \geq 0$$

$$\Leftrightarrow \left(\sum_{p \in \mathbb{C} \setminus \{0\}} \operatorname{ord}_p \cdot [p] \right) + \left(\operatorname{ord}_{p=0}(f) + 1 \right) \cdot [0] + \operatorname{ord}_{p=\infty}(f) \cdot [\infty] \geq 0$$

$\rightsquigarrow$ (f may have a pole of order ≤ 1
f has no pole at $[\infty]$

$\rightsquigarrow$ $f(z) = P(z)/z$ with $P(z) \in \mathbb{C}[X]_0$

$$P(z) := a_0 + a_1 z^1 + \cdots + a_{\deg P} z^{\deg P}$$

$$P\left(\frac{1}{w}\right) = a_0 + \cdots + a_{\deg P} w^{-\deg P}$$

$\therefore \operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(w P(1/w)\right) \Rightarrow \deg P \leq 1.$

$= 1 - \deg P \geq 0$

Let $P(z) = a + bz$ $(a, b \in \mathbb{C})$ $\therefore L(D) = \operatorname{span}\{1, 1/z\}$

$\therefore f(z) = \frac{P(z)}{z} = a + b/z$ $\quad \ell(D) = 2.$

6

- (3) $D = [0] + [1]$

allowed pole at 0 and 1

$\rightsquigarrow L(D) = \mathrm{span}_{\mathbb{C}}\{\frac{1}{z}, \frac{1}{z-1}, 1\}$

$\ell(D) = 3.$

- (4) $D = [\infty]$

allowed pole at ∞ $\rightsquigarrow L(D) = \mathrm{span}_{\mathbb{C}}\{z, 1\}$

$\ell(D) = 2$

- (5) $D = 2[\infty] \rightsquigarrow \begin{cases} L(D) = \mathrm{span}\{z^2, z, 1\} \\ \ell(D) = 3 \end{cases}$

- (6) $D = [0] + [\infty] \rightsquigarrow \begin{cases} L(D) = \mathrm{span}\{\frac{1}{z}, z, 1\} \\ \ell(D) = 3 \end{cases}$

- (7) $D = [\infty] - [1]$

$\mathrm{div}(f) + [\infty] - [1]$

$= \left(\sum_{p \in \mathbb{C}\setminus\{1\}} \mathrm{ord}_p(f)[p]\right) + (\mathrm{ord}_1(f) - 1)[1]$

$+ (\mathrm{ord}_\infty(f) + 1)[\infty] \geq 0$

7

$\therefore$ $\begin{cases} \operatorname{ord}_{p\notin\{1,\infty\}}(f)\geq 0 \\ \operatorname{ord}_{p=1}(f)\geq 1 \\ \operatorname{ord}_{\infty}(f)\geq -1 \end{cases} \Rightarrow f(z)=z-1$

$\rightarrow f\left(\frac{1}{w}\right)=\frac{1}{w}-1=\frac{(-w}{w}$

$\therefore L([\infty]-[1]) = \operatorname{span}_{\mathbb{C}}\{z-1\}$,

$\ell(D)=1$

- (B) $D=2[\infty]-[1]$

$\operatorname{div}(f)+2[\infty]-[1]$

$= \sum_{p\in\mathbb{C}\setminus\{1\}} \operatorname{ord}_p(f)[p] + (\operatorname{ord}_1(f)-1)[1]$

$+ (\operatorname{ord}_\infty(f)+2)[\infty]\geq 0$

$\therefore$ $\begin{cases} \operatorname{ord}_{p\notin\{1,\infty\}}(f)\geq 0 \\ \operatorname{ord}_1(f)\geq 1 \\ \operatorname{ord}_\infty(f)\geq 2 \end{cases} \Rightarrow f(z)=z-1$

$f\left(\frac{1}{w}\right)=\frac{w-1}{w}$

$\frac{w-1}{w^2}=\frac{1}{w}\left(\frac{w-1}{w}\right)=z(z-1)$ $\therefore L(D)=\operatorname{span}\{z-1, z(z-1)\}$

$\ell(D)=2.$

- (9) $D = 2[\infty] - [0] - [1]$

$$\deg(f) + 2[\infty] - [0] - [1] \geq 0$$

$$\sum_{p \in \mathbb{C}\setminus\{0,1\}} \operatorname{ord}_p(f)[p] + (\operatorname{ord}_0(f)-1)[0] + (\operatorname{ord}_1(f)-1)[1] + (\operatorname{ord}_\infty(f)+2)[\infty] \geq 0.$$

$$\therefore \begin{cases} \operatorname{ord}_{p \notin \{0,1,\infty\}}(f) \geq 0 \\ \operatorname{ord}_0(f) \geq 1 \\ \operatorname{ord}_1(f) \geq 1 \\ \operatorname{ord}_\infty(f) \geq -2 \end{cases} \Rightarrow f(z) = z(z-1)$$

$$\Rightarrow f\left(\frac{1}{w}\right) = \frac{1}{w}\left(\frac{1-w}{w}\right) = \frac{1-w}{w^2}$$

$$\therefore L(2[\infty] - [0] - [1]) = \operatorname{span}_{\mathbb{C}}\{z(z-1)\}$$

$$\ell(2[\infty] - [0] - [1]) = 1$$

- (10) $D = [0] - [\infty]$

$$\deg(f) + [0] - [\infty] \geq 0$$

$$\sum_{p \in \mathbb{C}\setminus\{0\}} \operatorname{ord}_p(f)[p] + (\operatorname{ord}_0(f)+1)[0] + (\operatorname{ord}_\infty(f)-1)[\infty] \geq 0$$

$$\therefore \begin{cases} \operatorname{ord}_{p \notin \{0,\infty\}}(f) \geq 0 \\ \operatorname{ord}_0(f) \geq -1 \\ \operatorname{ord}_\infty(f) \geq 1 \end{cases} \Rightarrow f(z) = \frac{1}{z}$$

$$\Rightarrow f\left(\frac{1}{w}\right) = w$$

$$\therefore L([0] - [\infty]) = \operatorname{span}_{\mathbb{C}}\{1/z\}, \quad \ell([0] - [\infty]) = 1.$$

Let $D = n[\infty]$ for some $n \in \mathbb{Z}$.

$$\operatorname{div}(f) + n[\infty] \geq 0$$

$$\sum_{p \in \mathbb{C}} \operatorname{ord}_p(f) + (\operatorname{ord}_\infty(f) + n)[\infty] \geq 0$$

$$\therefore \begin{cases} \operatorname{ord}_{p \neq \infty}(f) \geq 0 \\ \operatorname{ord}_\infty(f) \geq -n \end{cases}$$

(Case I) $n \geq 0$

$$\operatorname{ord}_\infty(f) \geq -n \rightsquigarrow f\left(\frac{1}{w}\right) = 1, w^{-1}, w^{-2}, \cdots w^{-n}$$

$$f(z) = 1, z, z^2, \cdots, z^n$$

$$\therefore \begin{cases} L(n[\infty]) = \operatorname{span}\{1, z, z^2, \cdots, z^n\} \\ \ell(n[\infty]) = n+1 \end{cases}$$

(Case II) $n < 0$.

$$\operatorname{ord}_\infty(f) \geq -n \rightsquigarrow f(z) = z^n$$

$$\rightarrow f \equiv 0$$

$$\therefore \begin{cases} L(n[\infty]) = 0 \\ \ell(n[\infty]) = 0 \end{cases}$$

10

$$\therefore \ell(n[\infty]) = \begin{cases} 0 & \text{if } n<0 \\ n+1 & \text{if } n\geq 0. \end{cases}$$

$$\Big\downarrow$$

$$D \sim (\deg D)[\infty] \text{ for all } D \in \mathrm{Div}(\mathbb{CP}^1)$$

$$\ell(D) = \begin{cases} 0 & \text{if } D<0 \\ \deg D + 1 & \text{if } D \geq 0 \end{cases}$$

$$D \sim D' \Rightarrow L(D) \simeq L(D')$$

Let $D \sim D'$ then

$$D = D' + \mathrm{div}(g) \text{ for some } g \in M(\mathbb{CP}^1)^\times$$

Let $f \in L(D)$ then

$$\mathrm{div}(f) + D \geq 0.$$

$$\mathrm{div}(gf) = \underbrace{\mathrm{div}(g)}_{+D'} + \underbrace{\mathrm{div}(f)}_{+D'} = \mathrm{div}(f) + D \geq 0$$

$$\therefore L(D) \simeq L(D') \qquad f \mapsto gf$$

$$\rightsquigarrow \ell(D) = \ell(D')$$